

Video 16: *Explicit Finite Difference Equations for Parabolic Equations*

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Unsteady Heat Conduction equation

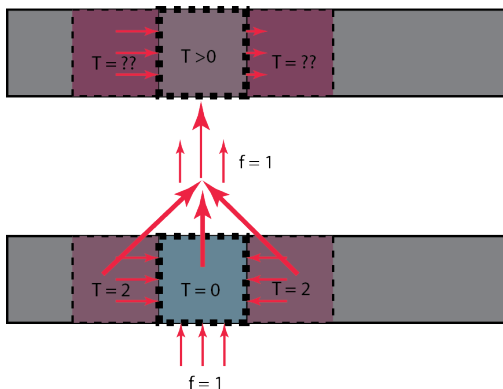
- Consider the unsteady heat conduction equation (parabolic equation):

$$\frac{\partial u}{\partial t} - \kappa \nabla^2 u = f \quad \text{in 1D for temperature} \rightarrow \frac{\partial T}{\partial t} - \kappa \frac{\partial^2 T}{\partial x^2} = f(x)$$

- What does this equation mean? Assume for now, u is temperature:
- The change of temperature w.r.t. t at a location x in a bar is due to:
 - Conduction/Spatial temperature differences:** The second derivative of the temperature w.r.t. the x -direction.
 - The temperature to the right and left of the point, x , will affect the temperature at x in future times.
 - Flux/Heating/cooling of the bar:** The heat flux, $f(x)$ into/out of the bar affects the temperature over time.

Unsteady Heat Conduction Equation

- Here is a pictorial view of the temperature in a bar problem:



- Temperature change due to conduction and heat flux.

Unsteady Heat Conduction equation

- How can we discretize this 1D unsteady heat transfer PDE using finite differences?

$$\frac{\partial u}{\partial t} - \kappa \nabla^2 u = f \quad (1)$$

- In time, we can discretize as a backward difference:

$$\frac{\partial u}{\partial t} \rightarrow \frac{u_{t,x} - u_{t-\Delta t,x}}{\Delta t} \quad (2)$$

- or as a forward difference:

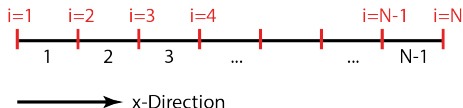
$$\frac{\partial u}{\partial t} \rightarrow \frac{u_{t+\Delta t,x} - u_{t,x}}{\Delta t} \quad (3)$$

- In space, we can discretize as:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} \rightarrow \frac{u_{t,x-\Delta x} - 2u_{t,x} + u_{t,x+\Delta x}}{\Delta x^2}$$

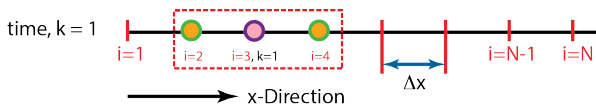
Unsteady Heat Conduction Equation: Discretize the Domain

- Divide the domain into $N - 1$ intervals of length Δx , with N nodes.



- We know how to discretize the 1-D spatial derivative, so let's start there – using the index- i :

$$\nabla^2 u_{t,i} = \frac{\partial^2 u_{t,i}}{\partial x^2} \rightarrow \frac{u_{t,i-1} - 2u_{t,i} + u_{t,i+1}}{\Delta x^2}$$

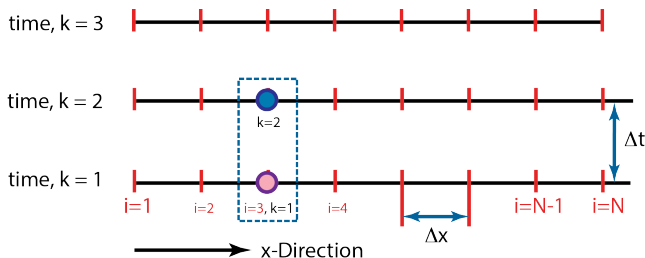


Unsteady Heat Conduction Equation: Time discretization

- Let's use the **forward derivative** in time.
- Let's divide time into Δt intervals represented by the k -index:

$$\frac{\partial u_{i,k}}{\partial t} \rightarrow \frac{u_{i,k+1} - u_{i,k}}{\Delta t}$$

- As an image:

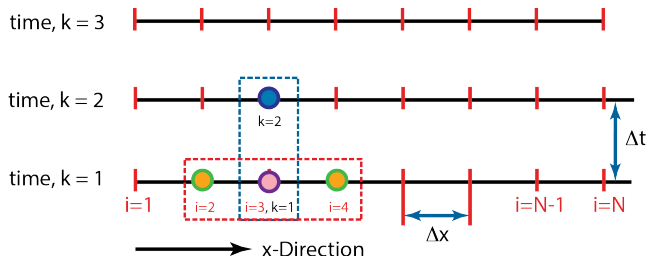


Overall Finite Difference Equation for a Node

- Here is the discrete equation with spatial and temporal derivatives combined :

$$\left(\frac{u_{i,k+1} - u_{i,k}}{\Delta t} \right) - \kappa \left(\frac{u_{i-1,k} - 2u_{i,k} + u_{i+1,k}}{\Delta x^2} \right) = f_{i,k}$$

- As an image, we can see the stencil:

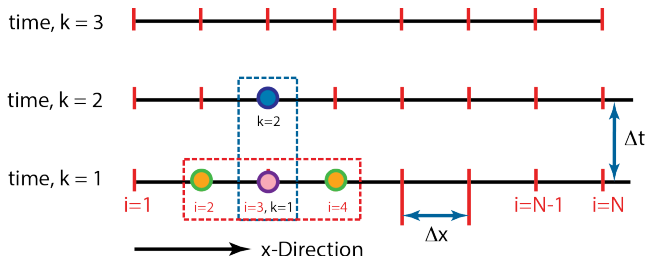


Overall Finite Difference Equation for a Node

- Re-arrange the equation – this is the familiar **forward Euler** method for ODEs:

$$u_{i,k+1} = u_{i,k} + \frac{\kappa \Delta t}{\Delta x^2} (u_{i-1,k} + 2u_{i,k} + u_{i+1,k}) + f_{i,k} \Delta t$$

- As an image, the second spatial derivative of temperature at x tells us how the temperature at x changes in time:

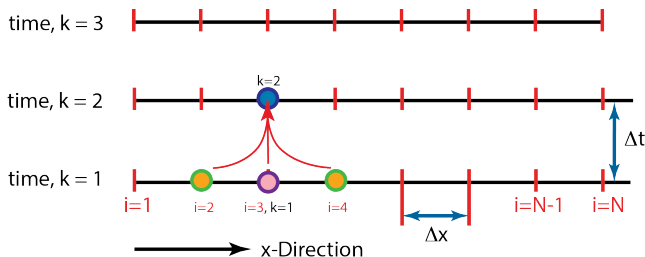


Nodal update equation

- The equation finds the new value of the temperature at $i, k + 1$:

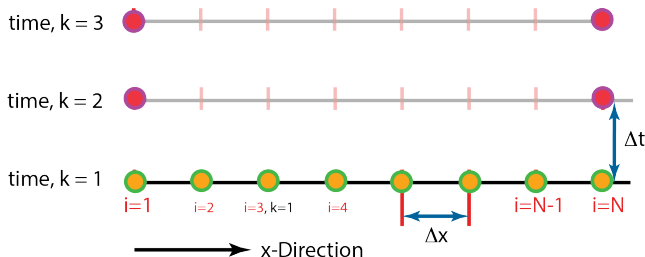
$$u_{i,k+1} = u_{i,k} + \frac{\kappa \Delta t}{\Delta x^2} (u_{i-1,k} - 2u_{i,k} + u_{i+1,k}) + f_{i,k} \Delta t$$

- As an image:



Setup initial conditions and boundary conditions

- The IVP (initial value problem) requires you know the solution/temperature at the initial time $\rightarrow$ initial condition (I.C.)
- The ends/boundaries of the bar also need boundary conditions (Dirichlet, Neumann, or other).



Psuedo Code and Matlab Example

```
1 - Setup your spatial discretization
--- deltaX = 1/(N-1)
2 - Setup your temporal discretization
--- deltat = dt; tEnd = Value;
3 - Setup the initial conditions in the bar
--- u(1, :) = Initial Condition in the bar
4 - Loop over time
--- While (k > tEnd)
4.1 - Loop over space
    for(i=2:N-1)
        u(i,k+1) = u(i,k) + kappa*deltat/(deltaX^2)*(u(i-1,k) - 2*u(i,k) + x(i+1,k)) + f*deltat
    end
4.2 - Set the boundary conditions
u(1,k) = Value\_LHS
u(end,k) = Value\_RHS
Done!
```

What have we learned

- Heat equation is a balance between time and spatial derivatives.
- The geometry and PDE discretization process for a parabolic equation is similar to elliptic equations.
- The solution process will be different though – notice there was no mention of matrices for this explicit solver.
- Need to march in one direction with the equation (to advance in time or one direction in space).
- This is called an explicit method (we use the current information to predict the future temperature) .